

# Omar Mahmoud

AI Engineer | Applied ML Scientist

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· Melbourne, VIC, Australia

## SUMMARY

Senior AI Engineer and Applied Scientist with a **PhD** in Trustworthy AI and 5+ years of hands-on experience building and deploying **large language model systems, ML pipelines, and NLP solutions** in both research and production environments. I specialise in taking ideas from experimentation through to real-world deployment, fine-tuning models, designing evaluation frameworks, and building reliable AI systems that work at scale.

My research focuses on **AI safety, privacy, and model alignment** with published work in top tier conferences. I bring a practical engineering mindset to research problems and a research depth to engineering challenges.

## TECHNICAL SKILLS

**LLMs & GenAI:** Fine-tuning (PEFT, LoRA, QLoRA) · RAG system design · Agentic AI workflows · LLM evaluation (RAGAS, TruLens) · Prompt engineering

**Trustworthy AI:** AI alignment · Privacy-preserving ML · LLM dememorisation · Hallucination mitigation · Adversarial robustness · Harmful content detection

**NLP & Speech:** Text classification · NER · Semantic search · Summarisation · Multilingual NLP · ASR (Whisper, wav2vec2)

**ML & Deep Learning:** PyTorch · TensorFlow · HuggingFace Transformers · Scikit-learn · Computer Vision · Experiment design

**Data & Retrieval:** Vector databases (FAISS, Pinecone, ChromaDB) · Elasticsearch · Experiment tracking (MLflow, W&B)

**Engineering & Cloud:** Python · FastAPI · Docker · AWS (S3, SageMaker, Lambda) · CI/CD · REST APIs · SQL

## EMPLOYMENT HISTORY

**AI Engineer** | *Deakin University (A2I2), Geelong, VIC*

Feb 2025 – Jun 2025

- **RAG system deployment:** Designed and shipped a production RAG system (LangChain, LlamaIndex) used by **500+ students and researchers**, combining hybrid retrieval with re-ranking and achieving a 20% accuracy improvement over baseline.
- **LLM fine-tuning & optimisation:** Fine-tuned open-source LLMs using PEFT/LoRA on domain corpora, reducing inference latency by 15% and improving task accuracy by 10% across classification and summarisation.
- **Safety evaluation:** Built LLM evaluation pipelines (RAGAS, TruLens) benchmarking hallucination, faithfulness, and alignment — establishing reusable protocols for responsible model deployment.
- **Privacy research:** Developed and evaluated dememorisation techniques to prevent LLMs from leaking training data, contributing findings to EMNLP 2023 and NAACL 2025.
- **MLOps:** Deployed and managed models via Docker and AWS SageMaker with experiment tracking and automated testing.

**Data Scientist** | *iNetworks — Cairo, Egypt*

Jan 2020 – Apr 2021

- **Recommendation system:** Built a real-time NLP-driven recommendation system using embeddings and text classification, increasing user engagement by 25%.
- **Data pipelines:** Engineered automated web scraping and preprocessing pipelines for structured and unstructured data, reducing manual effort by 40%.

**AI Engineer** | *Upwork — Remote*

May 2022 – Present

- **LLM & NLP delivery:** Designed and shipped 10+ production AI systems for clients across legal, healthcare, and e-commerce — including document Q&A, semantic search, and content classification pipelines on real-world multilingual data.
- **RAG & retrieval:** Built end-to-end RAG applications with custom chunking, embedding, and retrieval tuning (FAISS, Pinecone), reducing irrelevant results by ~30% over naive baselines.

- **Model optimisation:** Fine-tuned and deployed LLMs and NLP models via FastAPI on AWS; implemented monitoring, auth, and data-handling best practices for production reliability.
- **Speech & ASR:** Delivered ASR pipelines (Whisper, wav2vec2), achieving 15–20% WER improvement over off-the-shelf solutions on client audio data.

**Undergraduate Research Assistant** | *RSSCI Lab, Helwan University — Cairo, Egypt, Oct 2019 – 2021*

- **CV & NLP research:** Designed and executed experimental protocols for model evaluation across Computer Vision (COVID-19 chest X-ray detection — 1st place UGRF) and NLP tasks using PyTorch and TensorFlow.
- **Data engineering:** Led large-scale data acquisition and preprocessing for image and text datasets, ensuring integrity and feature readiness for model training.

## EDUCATION

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**Ph.D. Artificial Intelligence** | *Deakin University (A2I2), Melbourne, VIC* Sep 2023 – Jun 2026

Research: LLM alignment, privacy, memorisation, and multilingual NLP. Published at NAACL, EMNLP, EACL, and LREC.

**B.Sc. Computer Science** | *Helwan University, Cairo, Egypt*

Graduated Oct 2021

Faculty of Computers and Artificial Intelligence.

## SELECTED PUBLICATIONS

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**The Unintended Trade-Off of AI Alignment: Balancing Hallucination Mitigation and Safety in LLMs**

**EACL 2026** · Omar Mahmoud\*, Ali Khalil, Buddhika Laknath Semage, Thommen George Karimpanal, Santu Rana

**Aligning Multilingual Representations: Unveiling Multilingual Behavior Dynamics**

**LREC 2026** · Omar Mahmoud\*, Buddhika Laknath Semage, Thommen George Karimpanal, Santu Rana

**Alpaca Against Vicuna: Using LLMs to Uncover Memorization of LLMs**

**NAACL 2025** · Aly M. Kassem, Omar Mahmoud\*, Niloofar Mireshghallah, Hyunwoo Kim, Yulia Tsvetkov, Yejin Choi, Sherif Saad, Santu Rana

**Preserving Privacy Through Dememorization: An Unlearning Technique for Mitigating Memorization Risks in Language Models**

**EMNLP 2023** · Aly M. Kassem, Omar Mahmoud\*, Sherif Saad

**An Ensemble Transformer-Based Model for Arabic Sentiment Analysis**

**Social Network Analysis and Mining 2023** · Omar Mohamed\*, Aly M. Kassem, Ali Ashraf, Salma Jamal, Ensaf Hussein Mohamed

**GoF at Arabic Hate Speech 2022: Breaking the Loss Function Convention for Data-Imbalanced Arabic Offensive Text Detection** [1st place — Hate Speech Detection Shared Task (Sub-task A)]

**LREC 2022** · Aly Mostafa, Omar Mohamed Ahmed\*, Ali Ashraf

**GoF at Qur'an QA 2022: Efficient Question Answering for the Holy Qur'an Using Deep Learning** [3rd place — Qur'an QA 2022 Shared Task]

**LREC 2022** · Aly Mostafa, Omar Mohamed Ahmed\*

**On the Arabic Dialects' Identification: Overcoming Challenges of Geographical Similarities and Imbalanced Datasets**

**NADI 2022** · Salma Jamal, Aly Mostafa, Omar Mohamed Ahmed\*, Ali Ashraf

**COVID-19 Patient Chest X-Rays Automatic Detection Using Deep Learning** [1st place — 10th UGRF, Nile University]

**AMLTA 2021 (Springer)** · Aly Mostafa, Ahmed Elbehery, Ali Ashraf, Omar Mohamed Ahmed\*, Ali Mahmoud

**Arabic Speech Emotion Recognition Employing Wav2Vec2.0 and HuBERT**

**TMLAI 2021** · Omar Mohamed Ahmed\*, Salah Aly Ahmed [First Author]

**OCFormer: A Transformer-Based Model for Arabic Handwritten Text Recognition**

**IEEE MIUCC 2021** · Aly Mostafa, Omar Mohamed Ahmed\*, Ali Ashraf, Ahmed Elbehery, Salma Jamal, Ghada Khoriba, Amr S. Ghoneim

Pre-print: An End-to-End OCR Framework for Robust Arabic Handwriting Recognition (270M-word corpus).